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Introduction: Women, the Economy, and Economics

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The Oxford Handbook of Women and the Economy

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Abstract and Keywords

The transformation of women's lives in the past century is among the most significant and far-reaching social and economic phenomena, affecting not only women but also their partners and children. In developed and developing countries alike, women are acquiring more education, marrying later, having fewer children, and spending a far greater fraction of their adult lives in the labor force. This Handbook provides the first comprehensive collection of essays that addresses these issues, using the powerful framework of economics. The Introduction offers an overview of the field. It outlines its intellectual foundations and then reviews the many ways in which research into women's issues expanded and strengthened research methods and practices throughout economics. It then provides an overview of the thirty-one chapters in the Handbook. It concludes with thoughts about the strengths and weaknesses of the existing literature and about the further development of the field.

Keywords: women, marriage, labor supply, fertility, gender wage gap, household production

THE transformation of women's lives in the past century certainly ranks among the most significant and far-reaching social and economic phenomena, deeply affecting not only women themselves but also their partners and children. In developed and developing countries alike, women are acquiring more education, marrying later, having fewer children, and spending a far greater fraction of their adult lives in the labor force. Because women remain the primary caregivers of children, issues such as work/life balance and the glass ceiling remain at the forefront of policy discussions in the developed world. In developing countries, many women lack access to reproductive technology and are often relegated to jobs in the informal sector where pay is variable and job security is low or nonexistent. Considerable occupational segregation and stubborn gender pay gaps persist in labor markets around the world.

Many colleges and universities around the world now offer courses on women and the economy or on gender issues in economics, and textbooks covering the main topics do exist at the undergraduate level; see, for example, Blau and Winkler (2018) and Hoffman and Averett (2016). But there is no comparable collection at a more advanced level, nor any volume that features the leading scholars in the field. This Handbook provides the first such comprehensive collection of chapters that addresses these fascinating issues using the powerful analytical framework of economics. The Handbook chapters are divided into three broad sections—marriage and fertility; the labor market; and special topics and policy issues. These chapters provide outstanding examples of the broader use of economic theory to illuminate behavior, often outside of formal markets. Separate and special attention is given to developed and developing economies, because women often face different constraints depending on where they live. In each chapter, an acknowledged expert (or experts) in the field reviews the key trends, surveys the relevant economic theory, and then summarizes and, as appropriate, critiques the empirical research literature. The important issue of the identification of causal effects, as apart from correlations, is emphasized throughout, along with a clear-eyed view of what we know, what we do not know, and what critical unanswered questions remain. While the primary focus is on research in economics, where appropriate, the chapters incorporate the newest interdisciplinary research about gender from biology, psychology, and other disciplines. This Handbook should be of great value to a wide range of readers, from specialists to students, and also to scholars in other disciplines who want to see an overview of the most current research approaches concerning gender in economics.

In this introductory chapter, we first discuss the intellectual antecedents of the study of women in economics. How and why did women move from the periphery of economics to become one of its most studied and intellectually active research areas? Then we discuss the contribution that the study of women in economics made to the development of data and research methods that have enhanced research across almost areas of economics. Many methods routinely used in contemporary econometrics arose from the need to address specific problems posed by the analysis of women's economic behavior. We then provide an overview of the structure of the book and the core focus of each of its thirty-one chapters. We conclude with a discussion of the big issues remaining in this exciting area of inquiry.

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Modern economics is often thought to have begun with Adam Smith's *Wealth of Nations* in 1776. By that accounting, women were largely invisible within economics for a mere 175 years or so. Indeed, their activities were largely invisible in the recorded economy as well, since almost all their time was devoted to what we now understand and analyze as household production and thus was outside of standard measurements of economic

activity. They are invisible no more. This Handbook is emphatic evidence of that and of the rich contemporary scholarship in economics on women's issues.

Two parallel trends eventually moved women and women's lives into the mainstream of economic analysis. First, women began to transfer some of their time from household production to market work and thus became more visible in official statistics and other accounts of national economies. For example, at the beginning of the twentieth century, market work by women in the United States was essentially determined by marital status, race, and immigrant status. Virtually no white, nonimmigrant married women worked, as Collins and Moody (chapter 24) discuss in detail, and virtually no women of any description successfully combined work and family. By midcentury, this had changed modestly, and by the turn of the twenty-first century, it had changed unrecognizably. This kind of seismic change was true across multiple dimensions, including marriage, fertility, education, and occupation, to name just a few, and across all countries to a greater or lesser degree. All of these changes and their causes and consequences are discussed in this Handbook.

(p. 3) Second, the discipline of economics moved beyond its traditional focus on markets involving monetary exchange with formal prices to examine activities that involved resource use under scarcity but without the presence of formal markets and prices. The starting points were Lancaster's (1966) "A New Approach to Consumer Theory" and Becker's (1965) "A Theory of the Allocation of Time." While somewhat different in their approach, both involved reformulating demand theory in terms of the consumption of commodities that were the actual sources of utility and that were themselves the outputs of a production process involving market goods and time. That production process occurred outside of firms, primarily in households, and was performed predominantly by women. Interestingly, Lancaster never mentions women at all, and Becker, other than a few mentions in footnotes to empirical work by others, includes them only in a very odd context: "Women, the poor, children, the unemployed, etc., would be more willing to spend their time in a queue or otherwise ferreting out rationed goods than would high-earning men" (p. 516). But despite that conspicuous oversight, these papers opened the door to thinking more rigorously about activities outside the market sector using standard tools of economic analysis and thus made women's activities, including marriage, fertility, and other forms of household production, ripe analytical topics. The development of human capital theory by Schultz (1961) and Becker (1964) was another important stimulus to research on women. Formal education and on-the-job training were clearly important sources of human capital investment, but so too were families, especially mothers. We see the fruits of this expansion of the domain of economics throughout this Handbook.

Two other developments also made the analysis of women and the economy more productive. One was the development of longitudinal datasets, such as the Panel Study of Income Dynamics and National Longitudinal Surveys in the United States and a number of longitudinal household surveys launched in other countries including European countries, Australia, and Canada to name a few; this development is discussed more

thoroughly in the next section. These new datasets literally allowed economists to follow a woman as she moved through her life cycle, something that was impossible with cross-sectional data and which had been a critical issue complicating early analyses of women's labor force participation (Mincer 1962). Goldin (1983, 1990) had pioneered the use of synthetic life cycles by linking up ten-year birth cohorts across US censuses to outline how women's labor force activity varied in broad strokes over the life cycle. But these new longitudinal data sources provided much more information and facilitated more sophisticated micro-level analyses and the use of new econometric techniques.

Second, it also turned out that precisely because women's economic activity was in such flux over the twentieth century throughout the world as social norms were evolving to allow a wider range of opportunities for women, their decisions were both more varied and arguably more responsive to economic incentives. For example, since the early 1900s and in the wake of multiple technological, social, and economic transformations, the labor force participation rate of prime-age men (age 25 to 54) in the United States fell by less than 10 percentage points. Over the same time period, the (p. 4) corresponding participation rate for women increased by nearly 60 percentage points. To put it succinctly, women's economic behavior was both more varied and more interesting than men's. Explaining both cross-sectional differences and the time-series changes in women's behaviors was a fascinating challenge. Who worked and why? What were the roles of incomes, prices, and preferences? From a research perspective, there was not only more to explain but also potentially a greater role for economic analysis, because women were much more often on the margin of decision-making. The enormous changes in marriage and fertility rates were similarly tempting topics, as reflected in many of the chapters in this volume.

While economics and economists have provided a powerful analytical window into many women's issues, they have not always or consistently advanced the cause of gender equity. All economists learn about the efficiency of perfect competition, one implication of which is that in equilibrium, there are no unrealized opportunities for profit or gains to trade. It is a truly important and powerful result, but one limited to the existence of specific conditions that may not always exist in practice and also to an equilibrium situation that an economy is more often tending toward than actually residing in. Economists were, nevertheless, sometimes too ready to apply the implications of long-run competitive equilibrium to the status of women and the current state of gender inequities, concluding that the current state—whatever it was—was natural, inevitable, economically efficient, and therefore not usefully subject to change or policy intervention. If “greater gender equality was economically efficient,” they might argue, “it would already have been achieved,” because profitable opportunities are sought with monomaniacal zeal. Jacobsen (chapter 25) thoughtfully reminds us of these issues.

Becker, whose analytic contributions to the research literature on women's issues are unmatched by anyone, arguably did some of this. His model of taste-based discrimination famously concluded that gender pay differences unwarranted by productivity differences would disappear in appropriately competitive markets and thus focused much subsequent

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empirical research away from richer models of discrimination that might have fit gender issues more comfortably and toward finding productivity-based explanations of the gender pay gap.¹ Becker also constructed models that rationalized an extreme gender-based division of labor in households and occupational segregation, derived from underlying differences by gender in market and household productivity that were typically not themselves the subject of analysis. In one paper rationalizing the gender gap in earnings, he concluded that “married women spend less effort on each hour of market work than married men” and “economize on the effort expended on market work by seeking less demanding jobs” (Becker 1985, 533). He wrote this in the midst of the decade of the greatest change in the gender wage gap in the United States and as women were in the midst of a meteoric increase in their representation in graduate and professional schools.

Eighty years earlier, the 1900 US census had done something similar. After reporting on the major occupational categories for women—servants and waitresses, (p. 5) agricultural workers, dressmakers, laundresses, and teachers—the report noted that women were very lightly represented in the new and growing clerical and office work fields. Unhelpfully and stunningly incorrectly, the Census Bureau concluded, by the same kind of logic of inferring what will or should be from what is, that “many of these [clerical and copyist] occupations are not well adapted to the employment of women” (US Census Bureau 1907, 97). It is an easy error to commit: many of the chapters in this Handbook remind us that the many things that women do in their economic lives—from education and occupation to marriage and fertility—can and do change incredibly rapidly.

On a more personal note to the editors are the trends regarding women as PhD economists. Data from the Committee on the Status of Women in the Economics Profession (CSWEP) reveal that women remain underrepresented in the profession, although their numbers have increased substantially. In 1972, the first year that statistics on the status of women in the economics profession were collected, women accounted for only 6 percent of all faculty in economics departments, only 3 percent of full professors, and 12 percent of graduate students. By the mid-1990s, this percentage had increased to about one-third, and it has remained at that level ever since. The data make clear that, from entering PhD student to full professor, women have been and remain a minority in the economics profession. Within the tenure track, the higher the rank is, the lower the representation of women. In addition, women are far more likely than men to fall off the academic ladder at the time of promotion to tenured associate professor. This particular phenomenon appears to be unique to the economics profession, as reported in the CSWEP annual survey and report (2017). Women seem to be stuck at about 33 percent of all new PhDs—a number that has not changed in decades; even more discouraging is the fact that the percentage of undergraduate women who choose to major in economics is falling. And, just like at the top of the corporate ladder, women are particularly underrepresented at PhD-granting institutions compared to liberal arts colleges. Finally,

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in the more than 130-year history of the American Economics Association, only three women have served as president of the association.²

Interestingly, occupational segregation is even evident within the economics profession. The fields in economics that are represented in this Handbook are precisely those with the largest representation of women researchers within prominent economics departments (Dolado, Felgueroso, and Almunia 2012). Although women make up only 14.2 percent of faculty in the top fifty economics departments worldwide, they account for nearly one-quarter of all researchers in these departments in the field of health, education, and welfare. The next highest concentration of women in a field is labor economics (20 percent).

Finally, a systematic analysis of the comments on the website econjobrumors.org, a popular internet site for new economics PhDs seeking jobs, reveals derogatory comments disproportionately targeting women (Wu, 2017). This is suggestive of a hostile work environment still facing female economists.

(p. 6) Women and Economics: Data and Empirical Methods

The research on women featured in this Handbook would not have been possible without two parallel developments that transformed what kind of empirical work economists could do and how they could do it: the creation of new rich micro-data and the expansion of econometric methods that could be applied more easily due to advances in computing power. The two had a synergistic relationship: the new data allowed for more sophisticated analyses of, for example, women's life cycle labor force participation, which, in turn, raised new econometric issues and required new approaches.

The evolution of empirical work in economics, in many instances, has had its genesis in the fields of labor economics in general and family economics in particular. As Jacobsen (chapter 25) notes, "modern labor economics, including the 'new home economics,' dates from the early 1960s and only began to gain significant traction with the rise in affordable and accessible mainframe computing and systematic collection and dissemination of nationally representative, individual-level datasets, including panel data, in the mid- to late 1970s. This led to a rise in empirical analysis of female labor supply."

The availability of such data, including the original four cohorts of the National Longitudinal Surveys (NLS) and the intergenerational Panel Study of Income Dynamics (PSID) in the United States, provided fertile ground to apply panel data techniques to labor economics in general and the additional complications inherent in addressing gender issues in the labor market and in-home production. These data collection efforts, both begun in the 1960s, highlighted issues around gender, the labor market, and changing family and work responsibilities. In particular, the PSID, begun in 1967, was a family-based sample and adopted the then-standard convention of classifying the male in a married-couple family as the official family head for interviewing purposes. In 1976, the PSID recognized that this convention led to much poorer-quality data for wives, especially concerning work histories and earnings, and it fielded a separate wives' questionnaire that provided some of the earliest information on women's work histories.³ The initial cohorts of the National Longitudinal Surveys (older men, mature women, young men, and young women) consisted of separate surveys that focused on the particular issues faced by men and women in different age cohorts. These data included many respondents who could be linked by household identifiers to simultaneously observe married couples and in some cases their children as they progressed through the life course. As a reflection of the importance and greater variability of women's work and family choices during the decades of the 1970s and 1980s, the two female cohorts continued through 2003, while the two male samples were discontinued in 1981. New cohorts of the NLS, including both males and females, began in 1979 and 1997 and continue today.

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(p. 7) This longitudinal data that followed offspring allowed researchers to follow family members over their life course and link them together from generation to generation. Rather than using repeated cross-sections of data to infer the progress of successive cohorts, researchers were finally able to link decisions in early life to outcomes during the transition to adulthood and beyond.

The availability of data to examine issues of work and family was not exclusive to the United States. In the late 1970s, efforts began to harmonize sociodemographic and labor market data across middle- and high-income countries throughout the world. The Luxembourg Income Study now contains repeated cross-section micro-data on men and women in more than fifty countries. In addition, many countries gathered data on family behaviors in longitudinal household surveys such as the British Household Panel Survey, the Socio-Economic Panel in Germany, and the Household Income and Labor Dynamics in Australia. In developing countries, availability differs, but, beginning in the 1990s, most countries now administer demographic and health surveys that provide important insights into links between families, economic activity, and health.⁴

Empirical work that focused on issues of women's work and family contributions led to the development of new statistical techniques to deal with issues driven by gender differences in labor market behavior. An early example is Heckman's famous selection correction approach designed to address problems associated with the empirical analysis of women's labor supply (Heckman 1979). In the 1970s, female labor force participation in the United States hovered around 50 percent, posing a problem of missing data for the analysis of labor force participation. Economic theory emphasized the importance of wage rates in labor force participation and labor supply decisions, but researchers could observe wages only for working individuals. Naively imputing wages for nonworking individuals based on the observed characteristics of otherwise similar working women was clearly unsatisfactory. Heckman's technique allowed researchers to correct wage estimates for the likely selection bias due to unobservable differences between women who work and those who do not. The impact of this technique on empirical research in economics cannot be overstated. A Google Scholar search in 2018 for the terms "Heckman sample selection" yields well over one hundred thousand research papers that have used this technique, including in applications unrelated to gender and work. It was specifically cited as part of the basis for his Nobel Prize in Economics in 2000.⁵

Another important statistical method that evolved out of differences in labor market outcomes between men and women has become a standard part of the economist's empirical tool kit and is now used across fields in economics. The Oaxaca-Blinder decomposition (Oaxaca 1973; Blinder 1973) provides a method for partitioning the difference in group means into explained and unexplained portions. The initial application was the analysis of the gender wage gap, where the technique is used to decompose the overall gap into one component that results from differences in productivity (endowments and prior investments in human capital) and a second component due to differences in the compensation that men and women receive for identical productivity; these two

(p. 8) components are, respectively, the explained and unexplained shares (or residual

wage gap) of the wage gap. As of 2017, Oaxaca's original article has been cited more than seven thousand times, and a Google Scholar search for "Oaxaca decomposition" returns more than sixteen thousand results. In her chapter detailing the size and causes of the gender wage gap, Kunze (chapter 16) reports Oaxaca-style decomposition results from a large range of countries showing that differences in human capital characteristics (education and years of work experience) explain a sizeable portion, but less than half of the observed gender wage gap.

The hallmark of applied economic analysis is the focus on establishing causality by using statistical methods and strategic research design to combat problems of endogeneity. In this context, endogeneity can take a variety of forms, from sample selection, in which heterogeneous individuals select themselves into the treatment sample, to reverse causality and omitted variable bias. It is precisely this focused approach that has distinguished the empirical work of economists from others in the social sciences. Within economics, demographic and gender research has been an incubator for innovative empirical work to identify causality. To meet the challenge of designing empirical studies that can tease out causal relationships, economists have developed or adopted a number of experimental and statistical techniques to eliminate problems of endogeneity. Many of these techniques are highlighted in discussions of the empirical evidence applied to gender-specific topics in several of the chapters in this Handbook. A few of these examples are highlighted here.

Randomized controlled trials (RCTs) are considered the gold standard for addressing problems of endogeneity and thus establishing causality. By randomly assigning treatment to some participants (treatment group) but not to others (control group), the possibility of bias due to sample selection or omitted variables is eliminated. In her chapter on the biological bases of gender differences, Cobb-Clark (chapter 21) discusses the role of hormones as determinants of economic behavior. She describes an RCT study of postmenopausal women that finds no effect of randomly administered testosterone or estrogen on the results of numerous economic experiments designed to measure altruism, reciprocal fairness, trust, trustworthiness, and risk attitudes (Zethraeus et al. 2009). Opportunities for RCTs in social science research are more limited than in, for example, medical clinical research, but random assignment of treatment is often adopted to test the effectiveness of economic interventions in developing countries. The implementation of new programs, such as micro-credit programs, access to contraception, and income transfers to women, in randomly selected communities are discussed in chapters in this volume by Pörtner; Heath and Jayachandran; Menon and Rodgers; and Anukriti and Dasgupta.

Economists also conduct controlled experiments in the laboratory. Varying the constraints and/or options faced by participants allows the researcher to control for external variables affecting decision-making. In their chapter examining the impact of management practices on gender gaps in hiring and compensation, Kato and Kodama (chapter 23) highlight laboratory experiments designed to detect the presence of gender differences in preferences and decision-making. For instance, they discuss work by (p. 9)

Kuhn and Villeval (2015) in which participants are faced with real effort tasks in a laboratory setting that indicates that women are more likely than men to select a team compensation scheme as compared with compensation based on individual productivity.

One of the concerns about laboratory experiments is external validity. Can one set up a lab experiment in which the decisions required mirror those faced outside of the lab? Put differently, would an individual make the same decision when faced with the identical choices encountered in other settings? In an attempt to control the situations faced by research subjects outside of the context of a laboratory setting, field experiments are designed to impose randomization and control incentives and constraints in an externally valid setting. In this volume, Pan and Cortes and Shurchkov and Eckel both discuss research designed to answer the question “Are women less competitive than men?” In recent work, Flory, Leibbrandt, and List (2015) randomly vary the degree of competition in incentive schemes offered to job applicants and find that women are less likely to apply for a job with a competitive payment scheme. The researchers are able to control the description of the pay scheme that is offered to each applicant, but the context of decision-making is not one that is artificially described as part of laboratory decision-making, but rather in the actual context of a job interview. Similarly, the use of audit pairs in job-search settings can shed light on discriminatory hiring practices. In her chapter on gender pay gaps, Kunze (chapter 16) discusses studies in which fictitious job applications that are identical except for the gender of the applicant (Petit 2007) or motherhood status (Correll, Benard, and Paik 2007) are submitted to employers. These field studies simulate an RCT and can provide a causal estimate of the impact of these factors that could not be obtained from standard cross-sectional survey data.

In the absence of a researcher-designed randomized experiment, economists look for opportunities to exploit “natural experiments,” in which forces of nature or government policies mimic random exposure to a change in prices or other incentives. In this way, some individuals, distinguished by characteristics such as geography, age, or time, face different constraints. Appropriate statistical methods, such as difference-in-differences, instrumental variables, or regression discontinuity, can then be applied to tease out responses to the “exogenous” change. Many examples of the use of natural experiments in understanding gender-related socioeconomic and labor market outcomes are highlighted among the chapters in this Handbook.

Adserà and Ferrer (chapter 7) discuss studies examining parents’ simultaneous decisions concerning the quality and quantity of their children. Describing the use of an instrument for the quantity of children, they note that “[a] different way to measure the quality/quantity trade-off is through exogenous variation in the quantity of children induced by multiple births or by employing the sex composition of the first children as an instrument for fertility, under the assumption that parents aim for a balanced-gender composition.” Adserà and Ferrer discuss other studies using twin births and sex composition on educational investments in children in India (Rosenzweig and Wolpin 1980) and Norway (Black, Devereux, and Salvanes 2005).

A clever use of a natural experiment, discussed in the chapters by Lehrer and Son and Fletcher and Polos, concerns the impact of the media on teen fertility in the United States. The television series “16 and Pregnant,” which is about teen motherhood and is shown on MTV, a network targeting teens and young adults, has been identified as a vehicle for causing teens to think more realistically about the costs of an early birth. Using plausibly exogenous geographic variation in exposure to the airing of the show, Kearney and Levine (2015) find an increase in contraceptive inquiries via Google and Twitter and a corresponding decline in teen births.

Another statistical approach to identify causal relationships in research using natural experiments is difference-in-differences (DD). Under this approach, researchers compare behavior before and after the onset of the natural experiment for groups that were and were not subject to the treatment; the natural experiment essentially creates treatment and control groups. For example, Miller (chapter 22) describes a study that examines the impact of the gender composition of a board of directors on firm decision-making. The natural experiment occurred when Norway established a gender quota in 2006 for corporate leadership. In their study, Matsa and Miller (2013) use the board quota in public firms as a natural experiment that exogenously increased the proportion of women on the board, whereas private firms were not subject to the quota and served as a control group. Initially they compare the use of layoffs as a management strategy of firms before and after the externally imposed change in the number of women on governing boards. Adding another dimension of comparison, represented by a triple-differences model, they compare these pre- and postlayoff patterns of public firms (a control group not subject to the board quota) and private firms (the treatment group) with similar firms in other Nordic countries. They find that companies with more women on the board are less likely to lay off workers.

Instrumental Variable (IV) models can also be used to establish a causal relationship when endogeneity is a potential problem. A valid instrument provides exogenous variation in an explanatory variable without any independent effect on the dependent variable. In this way, the possibility of causality running in the reverse direction is eliminated. The chapters in this Handbook provide many examples of causal evidence provided by studies that employ IV approaches. As noted previously, natural experiments often serve as high-quality instruments. To determine the impact of sex ratios (the ratio of men to women) on a variety of family and labor market outcomes, researchers searching for exogenous variation in sex ratios have examined periods of high incarceration of men in the United States (Charles and Luoh 2010), war casualties in Europe (Abramitzky, Delavande, and Vasconcelos 2011), and enslavement of men in Africa (Teso 2016). These studies suggest causal impacts of declining sex ratios in lowering female marriage rates, increasing female education and labor force participation, and falling fertility rates (see Grossbard, Mansour and McKinnish, and Giuliano, this volume, for details of the findings of these studies).

Depending on the nature of the policy change, the impact of some natural experiments may be assessed by using a regression discontinuity approach. This type of analysis is appropriate when the control and treatment groups are distinguished by a discrete cutoff point along an observable continuous measure. For example, Guldi and (p. 11) Schmidt (chapter 19) examine whether public insurance programs adversely affect labor supply. They discuss work by Dague, DeLeire, and Leininger (2017) that examines the effect of a sudden imposition of an enrollment cap for Medicaid eligibility on the work behavior of childless adults in Wisconsin. Using a regression discontinuity framework, they show that individuals enrolled just prior to the imposition of the cap are less likely to be working than comparable individuals enrolled shortly thereafter.

Sometimes the best that researchers can do is to rely on temporal ordering using longitudinal data. With the availability of longitudinal micro-data at the individual or firm level, fixed-effects models are widely used to address issues of endogeneity such as sample selection or unobserved heterogeneity. Time-invariant individual characteristics are controlled by the inclusion of fixed effects, and such models allow researchers to examine the impact of changes in circumstances on individual behaviors. Describing a study using fixed effects at the individual level, Blau and Winkler (chapter 17) describe evidence that motherhood impacts women's wages. Endogeneity is a concern because lower wages among mothers might simply reflect selection into motherhood since women with lower wage offers will have lower costs of children. To address this specific concern, the common approach taken in this research is to estimate fixed-effect regressions. As Blau and Winkler report, Budig (2014) uses fixed effects to control for time-invariant individual-level factors in her study on fatherhood bonuses and motherhood penalties, finding evidence of a motherhood penalty but a wage premium for fathers.

Similarly, Johnson, Ribar, and Zhu (chapter 32) explore the relationship between homelessness and drug use. They discuss work by McVicar, Moschion, and van Ours (2015) in which "they used longitudinal data to consider the temporal ordering of these outcomes and used fixed-effects methods to control for unobservable individual characteristics that might contribute to both homelessness and substance abuse."

As noted by Antman (chapter 29), "It is important to keep in mind that the [credibility of the] results of any empirical study will often hinge on the particular identification strategy." One of the objectives of this Handbook is to include discussions of the identification strategy and credibility in establishing causal relationships in recent empirical work by economists that shed light on important gender-focused questions. These techniques are highlighted throughout the many chapters in this book.

Structure of the Book

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Some readers might wonder whether a volume like this is necessary today. After all, over the past fifty years, there has been a dramatic decline in gender disparities in key labor market outcomes including wages, participation rates, and occupational distributions in nearly all developed countries and in many developing countries. However, as each author in this Handbook makes clear, important gender disparities in labor market outcomes persist, and many are linked to women's family obligations.

(p. 12) This book is not a textbook per se. Each chapter can and does stand alone, although there are many valuable linkages and synergies among them and each includes cross-references to related chapters to aid readers in identifying associated material. In addition, readers will notice that many of the topics in one chapter also appear in another. This highlights the fact that decisions about marriage, fertility, education, and work are inextricably linked, particularly for women.

We divide this Handbook into three primary sections. Section I focuses on marriage and fertility. Section II focuses on women in the labor market, while section III is a compilation of chapters that address a variety of important topics and policies related to women and the economy.

Where necessary and/or appropriate, each chapter provides an overview of the key trends related to the outcome at hand, reviews the relevant economic theories that explain the underlying choices made by women, and then evaluates the empirical work testing these theories. Chapter authors have been careful to highlight studies with credible identification strategies. Each chapter suggests areas where future work is necessary. Because these chapters showcase the innovative empirical techniques discussed previously that are used to identify causal relationships, they present the most current thinking on the issues. Therefore, readers interested in doing research in these areas should find chapters in this Handbook to be a useful summary of the "state of the literature" and a helpful starting point for their own work.

Section I begins with marriage. Historically, marriage marked the beginning of adult life for women. Almost universally, women left their family homes once they married, usually ceased any formal attachment to the labor force, and in many cases/countries changed their names, thereby altering their identities. As economic circumstances have changed, patterns of marriage have changed. These changes have a strong educational gradient, as is evidenced in the next chapter.

Hani Mansour and Terra McKinnish focus their chapter on examining how marital sorting—who marries whom—has changed over time. Marital sorting is assumed to maximize the gains to marriage subject to the costs of marital search. Starting with a simple Becker model that predicts negative assortative matching on wages to maximize marital surplus, Mansour and McKinnish discuss theories that predict positive assortative matching and review the related empirical research. While much has been made of an apparent increase in positive assortative matching in the marriage market and its subsequent effects on income inequality, Mansour and McKinnish caution us that changes in the gains from marriage and the ways in which people meet prospective spouses are likely to

change the degree of marital sorting. Whether these changes will decrease or increase the degree of assortative mating on characteristics such as age, education, or race/ethnicity depends on the relative importance of preferences versus search costs.

Shoshana Grossbard's chapter reviews models of marriage, with special emphasis on how variations in the sex ratio can help explain outcomes such as marriage formation, the intramarriage distribution of consumption goods, labor supply, savings, type of relationship, divorce, and intermarriage. Importantly from a research perspective, she (p. 13) notes that changes in sex ratios can have profound impacts on outcomes in the marriage market. As one example of this, she emphasizes how the distribution of marital output and decision-making authority will change when sex ratios are unbalanced, with the group in excess supply faring more poorly. Unbalanced sex ratios in particular markets delineated by ethnicity, race, or education may also cause individuals to look outside of their traditional marriage market to find a match (Chiswick and Houseworth 2011).

No discussion of the economics of marriage is complete without a discussion of marital dissolution. Evelyn L. Lehrer and Yeon Jeong Son address the economics of divorce, with a focus on child and family well-being. Divorce is clearly endogenous to well-being, and many of the issues that scholars grapple with here require the econometric techniques described in the previous section of this introduction. A discussion of the relatively new "gray divorce" revolution occurring in many developed countries provides a window into what the future may hold for women, as they live longer and spend more of their adult lives working and perhaps less of them partnered.

While the first three chapters are largely focused on developed countries, most of the world's women live in developing countries. The distinctive features of marriage markets in developing countries are reviewed by S Anukriti and Shatanjaya Dasgupta in their chapter on marriage and families in the context of developing countries. In particular, they discuss two features of marriage markets that are today essentially nonexistent in developed countries—payments at the time of marriage (dowry and bride price) and polygyny. They pay special attention to the consequences of marital payments on socioeconomic outcomes. Anukriti and Dasgupta also discuss arranged marriages, which are much more common in the developing world although occasionally seen in parts of the developed world. They frame a discussion of assortative matching in a simple economic model that illustrates the conditions under which positive matching will occur and present the consequences of such matching, including the relative bargaining power of the partners. As their chapter makes clear, merely extending existing models of marriage that apply to developed countries will not capture the intricacies of spousal interactions in developing countries nor their economic consequences.

The next set of chapters focus on fertility issues. Fertility has been declining around the globe and in some developed countries hovers well below replacement level. In addition,

in many countries, the proportion of births that are nonmarital is rising. These important changes and their causes and consequences are discussed in the next set of chapters.

Claus C. Pörtner starts with a discussion of declining fertility in the developing world and addresses four provocative questions: (1) Why have fertility rates in Sub-Saharan Africa not fallen as rapidly as those in other parts of the developing world? (2) What factors determine the timing of fertility, and how is timing related to schooling and labor market outcomes? (3) What is the role of bargaining power in determining fertility? (4) How do sex preferences affect fertility outcomes? He also discusses efforts at family planning in developing countries, noting the complexities that are rooted in evaluating the success of these policies—the experiments are often small in scale, which makes it more difficult to establish whether an effect exists due to lack of statistical power, and (p. 14) noncompliance with randomization can be a problem. This can be particularly salient for family planning programs where the randomization often occurs at the community level, making it hard to avoid spill-over effects of, for example, information about contraceptives to nontreatment areas. Similar to the chapter on marriage in developing countries by Anukriti and Dasgupta, Pörtner's chapter illustrates the complexities that underlie fertility decisions in low-income countries and the heterogeneity across these countries that affects the outcomes we observe.

Fertility in developed countries is the focus of the next three chapters. Low fertility, now often below replacement in developed countries, has become a primary concern to policymakers as they grapple with the labor market and pension/aging consequences of declining fertility.

Alicia Adserà and Ana Ferrer document current trends in childbearing behavior in developed countries—including large drops in fertility rates and delayed fertility—and review some of the mechanisms that can explain them. Ultimately, these trends are linked to shifts in couples' demands for children following increases in women's education and labor market attachment and changes in access to family planning. They also discuss the recent emergence of a positive gap between desired and actual fertility linked to adverse economic conditions and high housing costs, as well as barriers for women regarding the ability to combine family and work. The chapter closes with a discussion of patterns of fertility among immigrants and of recent fertility policy experiences in developed countries.

Leonard M. Lopoo and Kerri M. Raissian tackle the topic of policies designed to alter fertility in the developed world. Given the low birth rates in many developed countries, their focus is on pronatalist policies including child subsidies—often referred to as “baby bonuses”—which are essentially cash subsidies paid to families for having children. They also discuss family-friendly work policies including maternity leave and child care, as well as incentives built into tax codes that are designed to encourage births. Their careful review of the literature reveals that the efficacy of these policies is quite mixed. In particular, the policies often affect the timing (tempo) of fertility but not the number of

children (quantum effect). These two chapters make clear that even after fertility rates decline, countries still wrestle with a host of issues, including the interconnectedness of the fertility and work/education decisions.

Digging deeper into fertility changes in developed countries is the task of Jason M. Fletcher and Jessica Polos, who focus on teen and nonmarital childbearing. The latter is on the rise in most developed countries and not surprisingly is linked to changes in marriage markets, discussed in the first two chapters. Both the causes and the consequences of nonmarital childbearing are examined. The consequences of teen and nonmarital childbearing have received a great deal of attention in the economics literature; here, the issue is whether the strong link between nonmarital childbearing and subsequent poor economic outcomes is causal or primarily correlational. This research is an excellent example of the way in which the analysis of women's issues raised complicated econometric problems and led to econometric innovations and highly creative research approaches.

(p. 15) Historically, the United States has had (and as of the writing of this chapter still has) the highest teen birth rate in the developed world. Yet, that rate has been falling as of late. One of the proximate causes of this decline is dramatic change in contraceptive technology. Martha J. Bailey and Jason M. Lindo, in their chapter, discuss how these changes have affected women's outcomes. Starting with the introduction of the birth control pill and legalized abortion in the 1960s and 1970s and continuing through to today with the introduction of long-acting reversible contraceptives (LARCs), they assess the impact of these technological changes on women's education and labor market outcomes. They provide an up-to-date survey of this literature and highlight many places where more work needs to be done.

In the next chapter, Elaina Rose weaves together a fascinating look at child gender and the family that bridges fertility in both developed and developing countries. While we often think of child gender as an issue that is most salient in developing countries, many of which exhibit a strong son preference, she makes clear that this is not unique to the developing world and discusses the myriad ways in which child gender affects parental behavior.

Kasey S. Buckles examines a topic of particular concern in the United States, as well as many other developed countries: fertility rates have fallen among the most educated and hence are higher for women of lower socioeconomic status (SES). A rich literature in economics and the social sciences has shown that improvements in women's SES can also improve the well-being of their children. This chapter identifies several channels for this effect, drawing on both theoretical and empirical work.

The last two chapters in this first section of this Handbook focus on two timely policy issues surrounding childbearing—child care and maternity and family leave. Jean Kimmel and Rachel Connelly tackle the convoluted child care policy in the United States,

providing a historical look at what they argue is a patchwork of policies that often fail to serve those who need them the most. They then propose an alternative to current US child care policy, one that they assert benefits all stakeholders and enhances child well-being.

Maya Rossin-Slater turns the focus to maternity and family leave policies. She skillfully summarizes decades of research that aims to identify the causal effect of policies in various countries on women's labor market outcomes and child well-being. Like the fertility policies highlighted in the chapter by Lopoo and Raissian, she finds that the efficacy of these policies varies substantially.

The inevitable conclusion of this first section of the book is that decisions regarding marriage and fertility are important determinants of economic well-being—indeed, they may matter more than labor market decisions.

This brings us to the second section of this Handbook: “Women in the Labor Market.” Here, the authors tackle several important related topics: women's labor force participation, employment, and earnings. Interspersed throughout these chapters are frequent references to fertility and marriage, which are inextricably tied to decisions regarding education, labor force participation, and earnings. Rachel Heath and Seema Jayachandran start this section with a chapter on women's education and work in developing countries. (p. 16) Two recent common trends in developing countries are an increase in female labor force participation and a narrowing of the gender gap in schooling. An important policy option discussed here is conditional cash transfers, which have been used with success to increase girls' education in many developing countries. Increased education and labor supply have prompted women to delay marriage and fertility and have improved children's health. However, there are also potential negative impacts, such as (perhaps paradoxically) an increased risk of intimate partner violence.

Astrid Kunze follows with a chapter on the gender wage gap in developed countries. She begins by noting that, despite narrowing, a gender wage gap persists in all countries. She takes the reader through the list of possible explanations for the existence of a wage gap (e.g., human capital, occupation choices, risk preferences, discrimination), carefully explains the two most common ways of decomposing a gender wage gap, and provides examples of the results of these decompositions.

Francine D. Blau and Anne E. Winkler examine the nexus between work and family. They begin by presenting a portrait of US women's labor market experience “then and now.” They focus on key challenges faced by women as they seek to combine motherhood and work: workforce interruptions due to childbearing, the impact of home and family responsibilities, and constraints posed by workplace culture and “how business is done.” They also draw attention to the very different experiences of women at the top and bottom of the educational distribution.

Despite the converging roles of men and women in the labor market, a strong common feature of labor markets around the world is occupational segregation. This segregation leads to wage differentials, as discussed by Patricia Cortes and Jessica Pan in their chapter on occupation and gender. While early work in this area attributed occupational segregation to gender differences in human capital accumulation or to discrimination being more pronounced in some occupations than others, more recent work has focused on risk preferences, attitudes toward competition, and women's preferences for family-friendly work environments. This evidence is reviewed in this chapter; they conclude their discussion with an intriguing look at policies that might lessen gender segregation in occupations.

Labor economists have long been interested in the way that taxes and transfers affect individual labor supply decisions. An application of this to women is found in the chapter by Melanie Guldi and Lucie Schmidt, who examine the effects of US tax and transfer programs on women's labor market choices. These programs are particularly important for women for three reasons. First, women are more likely to live in poverty and are therefore more likely to be eligible for means-tested programs. Additionally, women are more likely to live in families with children, which often triggers either eligibility for transfers or larger transfers. Second, in the United States, taxes are levied on the family as a unit and not at the individual level, which in turn influences women's labor supply choices, since they are still more likely than men to be the second earner in a household. Finally, women's labor supply has traditionally been more responsive to taxes and transfers than that of men, leading Goldin (2006) to argue that women "gave birth" to the modern study of labor supply.

(p. 17) In the next chapter, Olga Shurchkov and Catherine C. Eckel tackle the topic of how gender differences in preferences for competition, risk, and negotiation affect women's labor market outcomes, focusing on the rich literature in experimental economics. They report that experimental evidence confirms gender differences in these areas and that these might be linked to occupational choice and earnings. They also consider how gender differences in social preferences (defined as preferences for cooperation and nurturing) might affect outcomes.

Do well-known biological differences between men and women contribute to gender differences in labor market outcomes? Deborah A. Cobb-Clark examines this question in her chapter on biology and the labor market. She focuses her attention on four broad areas: (1) behavioral endocrinology, (2) human genetics, (3) neuroeconomics, and (4) sensory functioning and time-space perceptions. Contributions of the newly emerging subfield of neuroeconomics play an important role in measuring some of these differences. Similar to Shurchkov and Eckel, she concludes that while we can establish gender differences along these dimensions, linking them definitively to the gender wage gap, for example, is still an area ripe for research, as few datasets contain both good information on labor market outcomes and these important factors.

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The next two chapters address how women fare within firms. Despite decades of progress relative to men in work and schooling, women remain severely underrepresented among top corporate and political leaders. Amalia R. Miller discusses the current status and recent progress of women in leadership positions, focusing on corporate leadership. The effects of different policy interventions aimed at increasing female representation are assessed, with particular attention paid to gender quotas for corporate boards and to the question of spillover benefits from female leaders to other women. The effects of gender quotas in politics are also discussed and compared with those of gender quotas imposed on corporate governance boards. In a closely related topic, Takao Kato and Naomi Kodama peer inside the firm to see how women fare under various management practices. They review the high-performance work system (HPWS), as well as other management practices. Some of their findings may be surprising: for example, work-life balance practices that result in limited face-to-face interactions with coworkers may actually hamper women's career advancement. A rat race model, in which long working hours signal a worker's commitment, is a promising explanation for the gender gap in promotions. Corporate social responsibility practices may increase gender diversity.

While this book focuses on women and their economic lives, William J. Collins and Michael Q. Moody remind us in their chapter that the experiences of women are far from universal and an intersectional approach is imperative to understanding how different groups of women fared economically. Their focus is on African American women in the United States. They document and explore black-white differences in US women's labor force participation, occupations, and wages from 1940 to 2014. They draw on closely related research on selection into the labor force, discrimination, and pre-labor market characteristics, such as test scores, that are strongly associated with subsequent labor market outcomes.

(p. 18) While this Handbook is unabashedly focused on analysis through a neoclassical economics lens, we would be remiss if we did not include a chapter on feminist economics. Joyce P. Jacobsen, one of the academic leaders in the field, provides an overview of this perspective. She starts by outlining "the feminist economics intellectual project" with an eye to its intellectual development. Then she applies it to two important topics: labor supply and earnings and, of course, caring labor. Along the way, she highlights the differences between standard neoclassical theory and predictions, and feminist economic theory and predictions. This seems a fitting way to close the second part of the book.

Finally, section III consists of topical chapters that address issues related to women and the economy that are not neatly categorized as marriage/fertility or labor market, yet are clearly important for understanding women's role in the economy. The first chapter in this section is by Paola Giuliano, who leads us through the historical roots of gender differences. While we certainly all can identify gendered expectations for men and women, it is important to understand their origin. Reviewing work on the historical origin of differences in female labor force participation, fertility, education, marriage

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arrangements, competitive attitudes, domestic violence, and other forms of difference in gender norms, she illuminates how our past has shaped our present.

Much has been made recently about gender differences in longevity. Barbara Schone, in her chapter on gender differences in health, tackles this timely topic. There are important differences in the health of men and women. While women typically report worse health than men and suffer from more health conditions, they also live longer. These patterns become less paradoxical when analyzed in greater detail: women and men suffer from different conditions, especially at younger ages. In particular, women are more likely to suffer from chronic conditions, while men are more likely to suffer from life-threatening conditions. Schone documents these differences and summarizes biological, economic, and social explanations of sex differences in health. Despite a large literature exploring sex differences in health, much remains to be learned about the interaction of biological, economic, and social factors and their effects on health.

Nidhiya Menon and Yana van der Meulen Rodgers shed light on the relationship between economic development and women's empowerment. They focus on the structural drivers and constraints associated with the transition of women from unremunerated or low-paid production to higher-value work in three important labor market domains: entrepreneurship, agriculture, and wage employment. The chapter closes with the links between gender equality and economic growth, concluding that promoting gender equality can be a "gender-smart" way to achieve sustained economic development.

Francisca M. Antman examines women and migration. A significant number of women migrate each year—most from lower-income countries to higher-income countries—and the overwhelming majority are searching for better-paid work, often to send remittances back to their families. For many countries, these remittances represent a sizeable share of the receiving countries' gross domestic product. She also touches on the important ways in which nonmigrant women are affected by the migration of their spouses and other family members.

(p. 19) It is well known that women are more likely to perform work, paid or unpaid, that involves caring for others. In her chapter, noted feminist economist Nancy Folbre explains why care work often imposes a financial penalty that contributes to gender inequality. Such work often involves more personal connection, emotional attachment, and moral commitment than other forms of work. It creates public and private benefits, and its value is difficult to measure. All of these factors place care providers at an economic disadvantage and create what Folbre refers to as a "care penalty."

Jobs in science, technology, engineering, and mathematics (STEM) are among the highest paid in most economies. Yet, women are underrepresented in such jobs. Taking a life cycle approach, Shulamit Kahn and Donna Ginther explore explanations for the underrepresentation of women in STEM. Their discussion of causes also leads to some suggested strategies to correct such imbalances in this labor market. Along the way, they

document the STEM premium in earnings, making clear the importance of breaking down barriers to entry for women in these fields.

The last chapter, by Guy Johnson, David C. Ribar, and Anna Zhu, presents international evidence regarding women's homelessness. There is very little academic literature on this subject and even less focused on women, despite the fact that women are a sizeable fraction of the homeless and are often homeless with children. They start by defining and measuring homelessness, noting factors that are unique to homeless women. They then provide a discussion of causes and consequences of women's homelessness, reviewing important studies that have sought to credibly identify the effect of homelessness on outcomes including food security and drug use. Policy responses are an important aspect of this chapter as cities grapple with how to best serve this population.

Women, the Economy, and Economics: Where Do We Go from Here?

While this book provides the first comprehensive examination of women's economic lives in both developed and developing countries, we are cognizant, and our chapter authors make abundantly clear, that there is still a great deal that we do not yet know. It is fair to say that we are still in the early stages of having a full understanding of women's economic lives and the remaining challenges are substantial. No sooner do we come to grips with a particular issue—the wage gap, occupational segregation, or nonmarital fertility—than it heads off in a new, equally interesting, and sometimes unexpected direction. In the next two sections of this introduction, one focused on developed countries and one focused on developing countries, we highlight both the current state of knowledge, broadly speaking, and then the places where there is a dearth of evidence. An exhaustive summary here is not necessary; we strongly encourage readers to turn to (p. 20) the individual chapters in this Handbook to gain a full sense of the issues and challenges scholars are currently facing.

Developed Countries

Women in developed countries have made tremendous economic progress. They have greatly increased their labor force participation over the last fifty years. Economic and social forces leading to these dramatic increases in female labor force participation, which have occurred to some degree in all developed countries, include women's increased educational attainment, higher wages, improvements in household technology and contraceptive technology, and structural shifts in the economy that gave rise to greater availability of market substitutes for household production. In addition, structural changes in the economy that moved toward occupations requiring less physical strength

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facilitated women's widespread entry into the labor force and across a broader range of occupations.

Changes in family structure have also been important drivers of this change in labor force participation. Women at all ages are less likely to be married than were their mothers and grandmothers. This retreat from marriage is in part due to women's increased investment in their own human capital, the wide dissemination of reproductive technology including the pill and legalized abortion, changing laws that make divorce easier, and better household technology that frees up time from housework. In some countries, subsidized child care and paid maternity/family leave have also facilitated the entry of women into the workforce.

In many ways, in the developed world, today's women "have it all," certainly relative to earlier cohorts. As women's labor force participation rate approaches that of men and as their educational attainment often exceeds men's, studies report a narrowing of the difference in time spent in housework and child care by wives and husbands. However, just as in the past, women's economic circumstances and their choices regarding marriage and fertility are still intertwined in a way that men's choices are not.

Having it all, however, differs significantly by socioeconomic status. This is reflected in the education gradient in marriage and fertility that exists in many countries. This gradient suggests further inequality down the road. It is often the case that more educated women have fewer children and invest more in them than do less educated women. These differential investments in children portend an increase in inequality across families by mother's education. Policy solutions to this remain elusive.

Traditional explanations for the gender earnings gaps emphasized human capital differences, family responsibilities that varied by gender, occupational segregation, and discrimination. Human capital differences in years of schooling and years of work experience have substantially narrowed; in many countries, including the United States, women now have more schooling than men. But at least two conspicuous gender differences related to human capital still exist. First, women are still substantially underrepresented in STEM fields. Because jobs in STEM fields are some of the highest paying (p. 21) in modern economies, women's underrepresentation here doubtless contributes to the persistent gender wage gap. Understanding what forces lead to this underrepresentation and designing effective policies to mitigate it are critical issues that many countries are still trying to address.

Second, at the top of the corporate hierarchy, where women and men often have identical human capital, women have yet to break into the highest levels of corporate management (the so-called "glass ceiling"). Women have made inroads into corporate jobs, but remain far less likely to be in leadership positions. As Miller notes, "Women make up 50 percent of the world's population and 40 percent of its labor market participants, but they are severely underrepresented among business and political leaders. Only 19 percent of firms have female top managers and 23 percent of seats in national parliaments are held by women" (this volume, p. 539). This leadership gap undoubtedly contributes to the gender

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wage gap, because corporate leaders are typically among the highest-paid positions in business. Of course, leadership is accumulated through years of work experience, and given that women have only recently caught up to men in terms of experience and education, some lag in leadership representation is to be expected. However, just as in STEM fields, the pipeline leaks at many levels. Thus, even stocking the pipeline is no guarantee that women will catch up to men in this dimension, in part because of the glass ceiling that limits women's advancement to the top echelon of management positions.

This glass ceiling may be a result of taste-based discrimination or statistical discrimination against women, but it may also reflect differences in preferences for various aspects of work, especially work-life balance. For example, a well-known study that followed graduates of the University of Chicago MBA program found that even when men and women graduated from the same prestigious business school, it was not long before a meaningful gender wage gap developed (Bertrand, Goldin, and Katz 2010). This gap is partly related to parenthood, which has a negative effect on women's earnings and career trajectories, but not on men's. It is also due to gender differences in niches within business professions: men tended to pursue finance jobs, while women chose marketing and other paths that were more family oriented (Bertrand et al. 2010).

This study was notable because it focused on men and women who had very similar training (from a single university program) and were likely equally ambitious. Yet they ended up taking very different paths after graduation. However, we cannot discount the role of subtle discrimination that may be more salient at the top of the corporate hierarchy. The pipeline of women is clearly well stocked if measured by the number of women who have attained MBAs; in the United States, for example, women have now reached parity with men. The fact that in the United States, the number of women who are chief executive officers at Fortune 500 companies hovers around six annually suggests that other subtle barriers keep women from advancing to the top. Measuring these barriers is difficult, and much of what we know is anecdotal. Yet, there is no shortage of advice in the popular media. Books such as *Lean In* exhort women to push harder to be heard in the corporate arena. Economists have contributed by documenting some factors that might be explanatory, and experimental economists have established that women (p. 22) appear to gain less from negotiation, have weaker preferences than men for risk and competition, and may be more sensitive to social cues. Yet, we have yet to develop theoretical frameworks to fully understand the lack of women at the top. We do know that providing quotas can help ensure that women reach the top, and there is evidence that firms with women leaders do better along some metrics (see Miller, chapter 22, for more discussion).

The continuing role of gender discrimination should not be minimized, although it is arguably not as overt as in the past. Measuring discrimination has been difficult. Decomposition models of the sort put forth by Oaxaca quantify explained and unexplained (residual) portions of the gender pay gap. It is well understood that the residual wage gap is, at best, only a rough proxy for discrimination. As women's measurable skills approach and exceed those of men, we are left with ever larger unexplained portions from these

methods. The increased awareness of the limitations of the regression approach to measure discrimination has started to shift the empirical work on this topic toward field experiments such as audit studies, which aim to compare outcomes in the same job for two individuals who are identical in all respects other than gender. This approach has provided arguably cleaner evidence on gender discrimination in hiring and represents a promising path for further research, although it has its own limitations (Azmat and Petrongolo 2014).

Given that occupational segregation has certainly declined and the typical barriers are no longer as salient, more recent research emphasizes the role of gender differences in psychological traits, preferences for nonpecuniary (family-friendly) job characteristics, personality traits, and skills. Ascertaining that such gender differences exist has been the work of experimental economists, and we now have good data that suggests that women are more risk averse, less competitive, and less likely to negotiate than are men. What we do not yet have is good, consistent evidence on how these characteristics translate into either gendered behavior in the labor market or wage differentials. We are even further from having any sense of their quantitative importance. Still, these are potentially fruitful avenues for further research.

Biological differences are also potentially important explanations for wage gaps and occupational segregation.⁶ The state of this research is remarkably similar to the experimental evidence about gender, risk-taking, and competitiveness: we know that men and women differ across many biological dimensions, but not whether these differences translate into meaningful labor market effects. For example, the RCT described earlier in this introduction found that levels of testosterone are important predictors of altruism, trust, and risk attitudes, among other outcomes. But whether these differences explain wage differentials is still an area ripe for research. Part of the difficulty in pursuing this line of research is that social science datasets with high-quality earnings and occupation information for representative samples rarely have information on measures such as competitiveness, risk aversion, and testosterone levels, while health datasets that have biological information are typically small and unrepresentative and lack high-quality earnings information. Hence, it has been difficult for researchers to translate these into explanations for earnings differentials. It is also not fully understood how much of the (p. 23) observed gender differences in tastes and preferences for particular job attributes is environmental and how much is biological. The answer to this question is critically important to policymakers and managers who strive to make the workplace more egalitarian to harness the talents of all employees.

One potentially useful exercise would be to carefully examine countries that have achieved greater gender equality in earnings. For example, Belgium and Hungary have made remarkable progress in closing their unadjusted gender wage gaps (gaps in these two countries have narrowed by over 70 percent) and Italy, Norway, and the United Kingdom have narrowed their gaps by over 30 percent over the same time frame (authors' calculation from the Organisation for Economic Cooperation and Development [OECD]). At the same time, the gender gap in the United States has been virtually

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unchanged since the early 1990s. Whether this narrowing of the gap in these European countries is due to family-friendly policies in the workplace, a larger portion of women working in the public sector where wages tend to be more egalitarian, or the selection of only high earners into the workforce has been the subject of a great many studies and is an important area of ongoing research (see, e.g., Christofides, Polycarpou, and Vrachimis 2013).

While women have increased the human capital they bring to the workplace, firms have also tried to make their management practices more women friendly. Kato and Kodama (chapter 23) note that firms have sought out management practices aimed at leveling the playing field between men and women to varying degrees of success. Work-life balance initiatives that allow women (and men) to have more flexibility regarding work often limit an individual's face-to-face interactions with coworkers, and this may, paradoxically, hamper their career advancement. However, a lack of a comprehensive theoretical framework for understanding how firms' management practices are expected to affect women's earnings and employment coupled with a lack of firm-level data hinders our efforts to fully understand what goes on in the "black box" that is the firm. As in many other areas of economics, understanding the nuances of what occurs in firms is an important area of future research.

What can we do about the need to balance work and family responsibilities, which still fall mainly on women? Women predominantly bear the burden of caregiving, and with the aging of the population in developed countries, women are often tasked with caring for elderly parents, as well as their own children. This comes at a price and it is not unusual for a woman to leave the workforce or scale back her hours or career aspirations to fulfill her caring responsibilities, thus contributing to gender gaps in employment and earnings. We still regularly take it for granted that women are more nurturing and thus are more suited for such caring labor. A Becker approach would contend that because women's earnings are lower, they are the logical (efficient) choice when it comes to specializing in household production. However, these explanations are no longer very satisfying, as traditional gender stereotypes have been eroded. It is worth taking a step back and asking where these gendered expectations of caregiving responsibilities came from. The chapter by Giuliano demonstrates that gender role differences tend to arise in response to specific historical or technological circumstances and can persist long (p. 24) after those conditions have shifted.⁷ Changing these norms is thus a challenging task for policymakers. Policies that encourage men to take leave around the birth of a child have been enacted with some success in some countries (see Rossin-Slater, chapter 14) but are far from universal.

Many developed countries have made great strides when it comes to policies aimed at helping families (women) balance work and family. Paid maternity leave (often quite generous) is the norm in Europe, but is conspicuously absent in the United States. In fact, Blau and Winkler (chapter 17) report that the gap in family-friendly policies between the United States and other countries is large and growing, with consequent effects for women's labor force participation rates. Female labor force participation in the United

States has declined relative to other economically developed countries, and they argue that this is due to the greater expansion in OECD countries of family-friendly policies, including the length and generosity of parental leave, mandates for part-time availability, and expenditures on child care. The chapter by Kimmel and Connelly in this volume makes clear that the United States is woefully behind in the provision of child care as well.

But providing maternity leave and child care that can aid women as they move forward in their careers is probably not sufficient to ensure gender equality in labor markets. The very nature of work has evolved in some occupations in ways that are decidedly not compatible with work-life balance, for either men or women. A famous study by Goldin (2014), which is discussed in several chapters in this volume (e.g., Blau and Winkler, Kunze, Cortes and Pan), examined why the gender gap for college-educated women in the United States varied across occupations by linking those gaps to inherent characteristics of the occupation, including the degree of autonomy and whether time flexibility was likely to be easy or difficult to accommodate in terms of production.⁸ For example, some occupations, such as law, business, or some medical specialties, may require extensive amounts of “face time” or client interaction or otherwise be structured so that different workers are poor substitutes for one another in production. As a result, these occupations may develop nonlinear pay schedules in which long hours are highly rewarded. Other jobs may have a more linear relationship between hours and pay, especially where technological change allows a job to be performed by two or more different persons at different times without loss of productivity. Goldin cites pharmacists as an example of this last category. The advent of medical records technology means that patients are not tied to a specific pharmacist; instead, pharmacists are near-perfect substitutes for one another. Consequently, hours worked by pharmacists are more flexible than in the past and the gender gap is among the smallest of any occupation. The tricky part here is to understand what can be done about jobs with nonlinear relationships between hours and pay. Goldin is optimistic that technological changes that facilitate off-site and off-hours communication may play a role. Restructuring client expectations is another possibility, but it is also possible that women are less likely to select these jobs either because they are less family friendly or because they require a more competitive nature. Finally, we cannot discount the idea that within certain occupations the expectation of long hours has become a social norm, even where it is (p. 25) not technologically required. These norms may create an unnecessary barrier as a form of subtle bias against working with women in these roles.

While the gender gap in schooling has closed—and even reversed—in most developed countries, there are remaining gender differences in pay and employment levels, as well as in the types of activities that men and women perform in the labor market. Understanding these persistent differences is the task of the next generation of labor and demographic economists. Closer collaboration between economists and both psychology

and medical researchers may be necessary as scholars turn to biological and social differences to fully understand remaining gaps.

Developing Countries

Despite falling marriage rates and rising age at first marriage, developing countries are still characterized by higher rates of marriage and earlier age at first marriage as compared to developed countries, which often translates into lower education for women and higher fertility, with all the attendant child and maternal health issues this high fertility involves. Because women are often less valued in these economies, authority within marriage is often unequal. Some of the most interesting research has focused on the interactions between marriage institutions and norms and women's education and fertility. In this volume, Anukriti and Dasgupta emphasize that these interactions have strong policy implications for the well-being of women. For example, the value of education for women may vary across groups that do and do not engage in bride price payments at marriage, because a more educated bride may command a higher bride price. Polygyny is still practiced in some developing countries, with its own implications for fertility and women's autonomy. The role of polygyny in economic growth is another issue for further study.

In part a reflection of these changes in marriage, fertility is falling across developing countries. Differences in preferences for children by husbands and wives raise issues related to bargaining power when explaining fertility declines in developing countries. As women gain bargaining power, it is often thought they will want fewer children, but the reality is likely to be more nuanced. As Pörtner makes clear in his chapter in this volume, the role of bargaining in fertility is still an area where researchers need to know more to better inform policymakers. Merely providing access to contraception may not be enough to reduce fertility unless the relative bargaining power of wives in the decision to use contraception is accounted for.

An especially important aspect of intrahousehold allocation is son preference, which remains important in China and India, as well as some other countries. Son preference can affect both the timing and spacing of births. A strong son preference can mean an imbalance in the sex ratio and is blamed for the phenomenon of "missing girls." Achieving gender equality is hampered when there are missing women. The widespread adoption of technology that allows couples to determine the sex of an unborn child, combined with lower desired fertility as in India or more formally as with China's (p. 26) One Child Policy, can paradoxically worsen the problem of missing girls as women lower their fertility rates but maintain their preference for sons. This leads to undesirable changes in the sex ratio that affect both marriage markets and the labor force. The relaxing of China's One Child Policy may do more to equalize the sex ratio than any gender-targeted policy can.

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One of the driving factors for lowered fertility has been improved opportunities for education for women in developing countries. Although the link between education and fertility is strong, the mechanisms are not well understood. Does education increase knowledge about contraception use or does the higher opportunity cost of children operate to reduce fertility? Furthermore, we do not know exactly how much education is enough to lower fertility. Is completing primary school sufficient or do girls need to get through high school? Pörtner (this volume) reports that one reason for persistently higher fertility in Sub-Saharan Africa is that it may take a higher level or better quality of education for girls before the effect on fertility is seen. Clearly, we need to know more about this important linkage.

Like women in developed countries, women in developing countries have primary family responsibilities and perform the bulk of unpaid care work. Lack of child care alternatives coupled with the need to gather basic supplies such as water and cooking fuel create time-consuming barriers to women's employment in developing countries. According to the World Bank's 2012 *World Development Report: Gender Equality and Development*, closing these gender gaps matters for economic development and policymaking (World Bank 2012). Greater gender equality can enhance economic productivity, improve development outcomes for the next generation, and make institutions and policies more representative.

Women in developing countries have also seen important changes in their working lives. Women have been identified by multinational corporations as cheap, productive, and reliable labor, particularly in the textile industry, and this has consequences for their bargaining power as economic development has meant more jobs in the formal sector. Yet, it remains the case that when women are in paid employment, they are more likely to be engaged in part-time rather than full-time work and in the informal rather than the formal sector, and across the globe women earn less than men for comparable work.

A lack of well-paying jobs in developing countries has often pushed women to migrate in search of work. As Antman (chapter 29) notes, women are both likely to migrate and to be left behind as their husbands migrate. Thus, understanding the impact of migration and remittances on households "left behind" in source countries is important as it has profound implications for economic development. Remittances from a spouse or other family members are a source of income for women left behind, and remittances from women who migrate can be an important source of revenue in their countries of origin.

Economic development has the potential to be advantageous for women; hence, it is important to understand their role in that development. To that end, economists and policymakers are interested in understanding how microcredit can lift women out of poverty by providing them small loans to start a business. Made famous by the Grameen Bank in Bangladesh, microcredit has swiftly moved across the developing world. Heath and Jaychandran (chapter 15) note that while microcredit is not necessarily a gendered policy, many organizations do target female entrepreneurs, who are known as being good borrowers, and repeated studies have shown that when money is in the hands

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of women, it is more likely to be invested in children. As Menon and Rodgers note in their chapter in this volume, such initiatives target individuals who have difficulty obtaining conventional loans through commercial banks. Women, in particular, have faced such difficulties due to their lack of collateral, a problem that is exacerbated by weak or nonexistent property rights for women in many developing countries.

Several RCTs have evaluated the efficacy of microcredit in terms of its ability to foster female entrepreneurship. These studies tend to find positive but small impacts; however, the precise impact on women's empowerment and entrepreneurship varies by program specifics and context. Other studies have noted that business skills training can be effective in the developing world to help women as they work toward self-employment, and land ownership rights to women result in improved economic sufficiency and increased overall productivity. Though promising, additional evaluation is necessary to understand if widespread gains are possible from these types of programs. It seems apparent that cultural context and patriarchal attitudes may moderate the effectiveness of these programs, but there remains work to be done to fully understand the effectiveness of these policies in varying contexts.

Finally, economists are studying the relationship between women's empowerment and economic development. Much has been made in the world of economic development of the strong correlation between women's empowerment and economic growth both within a country and across developing countries. Yet defining what it means to improve women's economic empowerment and move toward gender equality is difficult. For example, should we focus on the equality of opportunities or the equality of outcomes? Untangling causation from correlation in this context is difficult. Clearly economic growth itself has provided and will continue to provide more opportunities for women. As Menon and Rodgers (chapter 28) note, improving educational attainment for women and girls can strengthen the ability of household members to engage in productive activities and improve the efficacy of the labor force, thereby bolstering the economy's growth potential. Yet, without economic growth, there may not be schools and infrastructure to support women's increased schooling. And, given women's roles as caretakers, policies that facilitate entry into the labor force, ensure equality in earnings and access to employment, and provide safe workplaces are essential to harnessing the full potential of the world's women. Yet, it remains the case that in many developing countries women lack sexual and reproductive autonomy or the ability to own property or to obtain an education.

Understanding the impact of policies that aim to strengthen women's political and economic autonomy is an important ongoing goal, but one that faces challenges. Though researchers are often able to implement demonstration programs that include the collection of data from randomized studies, the impact of implementation on a larger scale and in different cultural contexts is often unknown.

(p. 28)

Conclusion

In recent years, social and economic forces have changed the landscape of both family and work life, and women are at the center of these changes. The chapters in this Handbook make clear that addressing the link between work and family is crucial to women's continued economic progress. In addition, promoting gender equality is essential for economic growth and for the well-being of the next generations.

Despite considerable progress, a number of gaps remain in our understanding of economic gender differences. As Anukriti and Dasgupta point out, "more rigorous theoretical and empirical work is needed to understand the implications of various cultural norms, customs, and practices for marital matching, intrahousehold decision making, and macroeconomic outcomes. . . . [S]uch a focus is also extremely policy relevant, and can help us understand the unique challenges and trade-offs involved with policymaking" (this volume, p. 114). As our expert authors have also noted, collecting better data will allow us to further refine our understanding of barriers that keep gender wage, employment, and occupation gaps from closing. Continuing to advance and test theories that can uncover the increasingly nuanced reasons for gender differences in behaviors and outcomes is also imperative. Having this information at hand is crucial for policymaking. Policymakers can only make informed policy when they have good data about which policies are effective. Many countries are motivated to develop policies to advance women's economic well-being. It is our hope that this Handbook will be an important resource for scholars, who will then provide the knowledge necessary to make effective policy choices in the future.

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Notes:

(1.) Becker's taste-based model was developed primarily to explain discrimination based on race and it is premised on a "dislike" of the group subject to discrimination (Becker 1957). That preference fits gender discrimination somewhat awkwardly. To be fair, the recognition of the existence of gender-based discrimination was still a decade or more in the future.

(2.) The first woman to serve as president of the American Economics Association was Alice Rivlin (1985), a well-known nonacademic economist. She was followed by Anne Krueger (1996) and Claudia Goldin (2013).

(3.) A well-known early use of this work-history data is Corcoran and Duncan (1979).

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(4.) See information on the Demographic and Health Surveys programs, with information about surveys across ninety countries, at <http://dhsprogram.com/Data/>.

(5.) The Nobel Committee cited Heckman “for his development of theory and methods for analyzing selective samples” (https://www.nobelprize.org/nobel_prizes/economic-sciences/laureates/2000/heckman-facts.html).

(6.) Biology also affects gender differences in well-being through another path. Differences in life expectancy across gender are also important. While women live longer than men in virtually all countries, life expectancy varies greatly by socioeconomic status, and women are generally less healthy compared to men. As Schone (this volume) notes, some of these differences may be biological and hence perhaps less mutable to policy, but others are clearly rooted in social and economic factors.

(7.) Alesina, Giuliano, and Nunn (2013) show this in the very specific context of plough versus shifting cultivation (slash and burn) agriculture, linking those technologies to differences in gender roles and even to current attitudes.

(8.) The source of the occupational characteristics data is O*Net, the US Department of Labor successor to the earlier Dictionary of Occupational Titles. This dataset is still relatively little used, but may be a promising avenue for future research.

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